EXPERIMENT NO. 05 – CLOCK DATA RECOVERY



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TITLE: Design of Clock Data Recovery (CDR) Circuits

PROBLEM STATEMENT: Using the PN sequence generated in the first experiment as data, design an elementary clock data recovery (CDR) circuit**.**

KEY COMPONENTS: IC CD4046**,** IC7486, IC74164, IC74LS74, Resistors, Capacitors, Digital oscilloscope, Function generator, Breadboard, Connecting wires, and Power supply.

BRIEF THEORY & BACKGROUND:

The experiment aims to design a circuit for the clock data recovery (CDR) at the receiver using the PN sequence generated. The clock generated should correspond to the one used to generate the PN sequence.

Clock and data recovery (CDR) is essential in high-speed transceivers to extract a synchronized clock from noisy, asynchronous data streams for reliable communication. It also retunes the data to remove accumulated jitter, posing significant design challenges in meeting stringent communication standards [3].

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As mentioned in [1], the clock signal must adhere to several key criteria:

1. Its frequency should match the data rate; for instance, a data rate of 10 Gb/s (where each bit lasts 100 ps) corresponds to a clock frequency of 10 GHz, featuring a 100 ps period.
2. It should maintain a precise phase relationship with the data stream to optimize the sampling of individual bits by the clock. When the rising edges of the clock align with the midpoint of each bit, it ensures that sampling occurs at a point furthest from preceding and subsequent data transitions, allowing for maximum tolerance to jitter and other timing uncertainties.
3. The clock signal should exhibit minimal jitter, as it constitutes the primary contributor to the jitter in the retimed data.

A CDR phase detector compares the phase of incoming data with the recovered clock to adjust the clock's sampling phase. There are two types of Phase detectors: Linear phase detectors (e.g., Hogge) provide both the magnitude and direction of phase error, while non-linear ones (e.g., Alexander, Bang-Bang, oversampling) give only the error's direction [5].

EXPECTED OUTCOMES:

Design and implement the following circuit to demonstrate clock data recovery. Note that several parts need to be carefully designed.

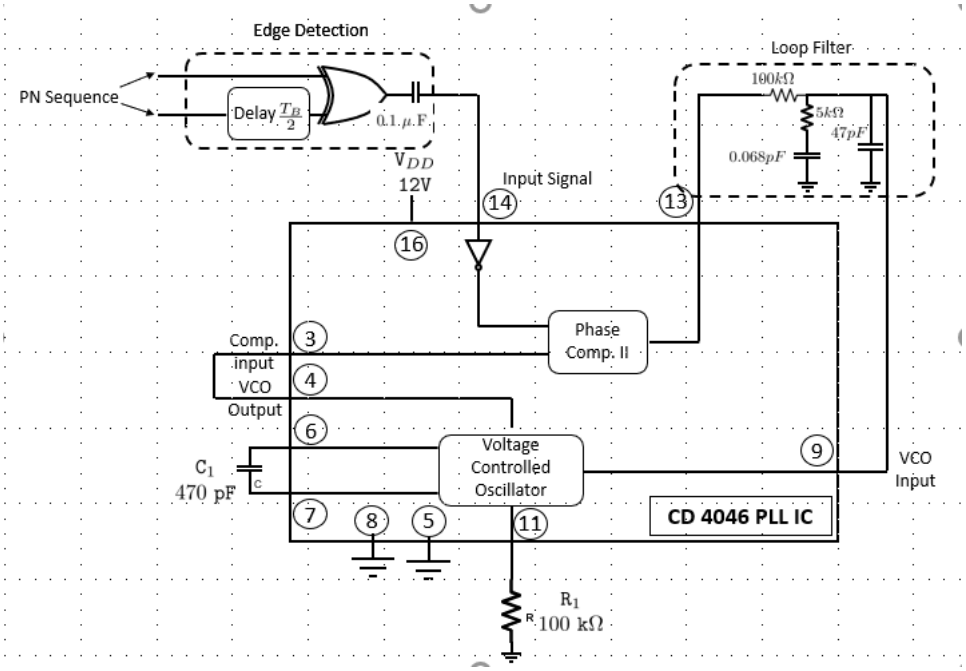


Fig. 1: Clock data recovery circuit [2]

CD 4046: Phase Locked Loop IC

The figure below shows a block diagram of the 16-pin CD4046B IC, which has been implemented on a single monolithic integrated circuit. The PLL structure consists of a low-power, linear VCO and two different phase comparators, having a common signal-input amplifier and a common comparator input. A 5.2-V Zener diode is provided for supply regulation, if necessary. The VCO can be connected either directly or through frequency dividers to the comparator input of the phase comparators. The LPF is implemented through external parts because of the radical configuration changes from application to application and because some of the components cannot be integrated [2].

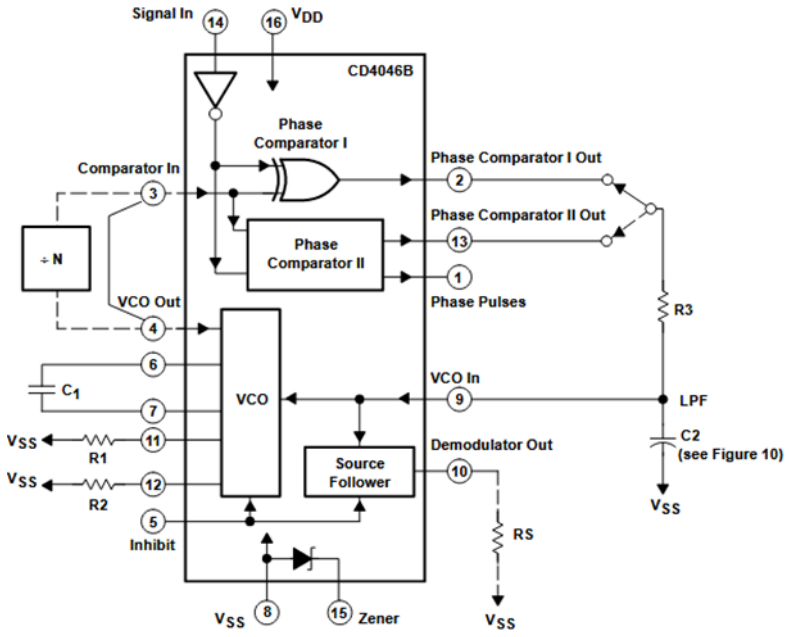


Fig. 2: CD 4046B PLL IC [2]

Coming back to the CDR design problem, there are several design sub-problems that need to be considered for the experiment.

1. **Edge detector:** From the input PN sequence (or data sequence in general) spikes are extracted [4].

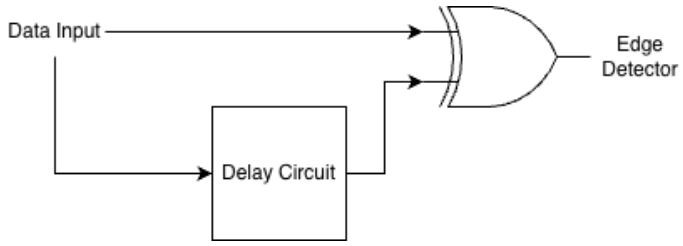


Fig 3: Edge Detector Circuit [4]

2. **Phase-locked Loop (PLL):** A basic PLL is given in the following block diagram:

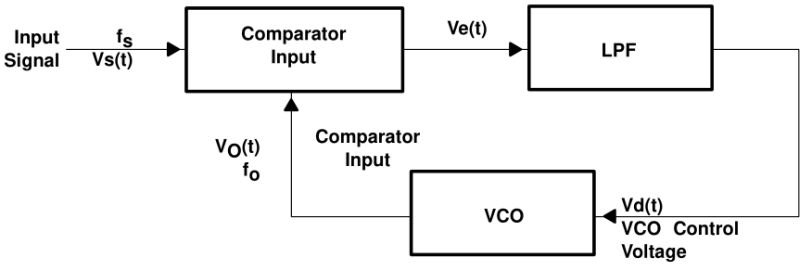


Fig 4: Basic PLL Block Diagram [2]

An important consideration for the design of the PLL is the feedback loop gain. Seen from the perspective of a closed-loop feedback control system***, stability*** of the filter is essential and this is crucially characterized through the filter.

3. **Jitter suppression**: This problem couples with the PLL design problem. Jitter can be a serious problem in the recovered clock [4].

Types of Jitter include

Long term Jitter

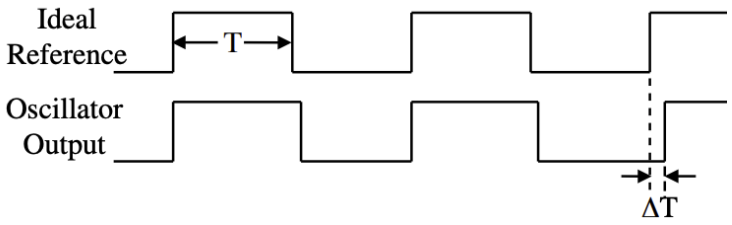


Fig 5: Long Term Jitter [4]

Cycle to cycle Jitter

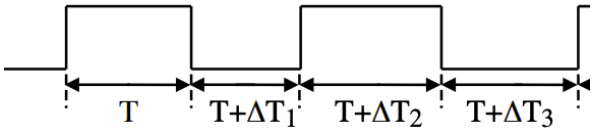


Fig 6: Cycle to Cycle Jitter [4]

Loop filter design considerations:

The loop filter design is crucial to ensure both stability of the feedback control system as well as to compensate for any jitter in the recovered clock.

PROBLEM SETUP ASSUMPTIONS:

Assume the following for the experiment:

Clock frequency for generating the PN sequence: 10kHz.

R_b=100kΩ, R_a=5 kΩ, C_a=0.068μF, C_b=47μF. These choices ensure that the VCO tracks a range of frequencies from 0Hz to 70kHz.

OUTCOMES:

1. Present the recovered clock signal and comment on the clock jitter present, if any, and comment on the potential weakness in the design for this jitter. How would you compensate for it?

2. Identify the specific assumptions made in the design of the CDR circuit (what is the receiver assumed to know so as to complete the design?) and point out the potential limitations of the circuit discussed.

3. Suggest improvements to the design while getting rid of the assumptions to make the circuit more realistic.

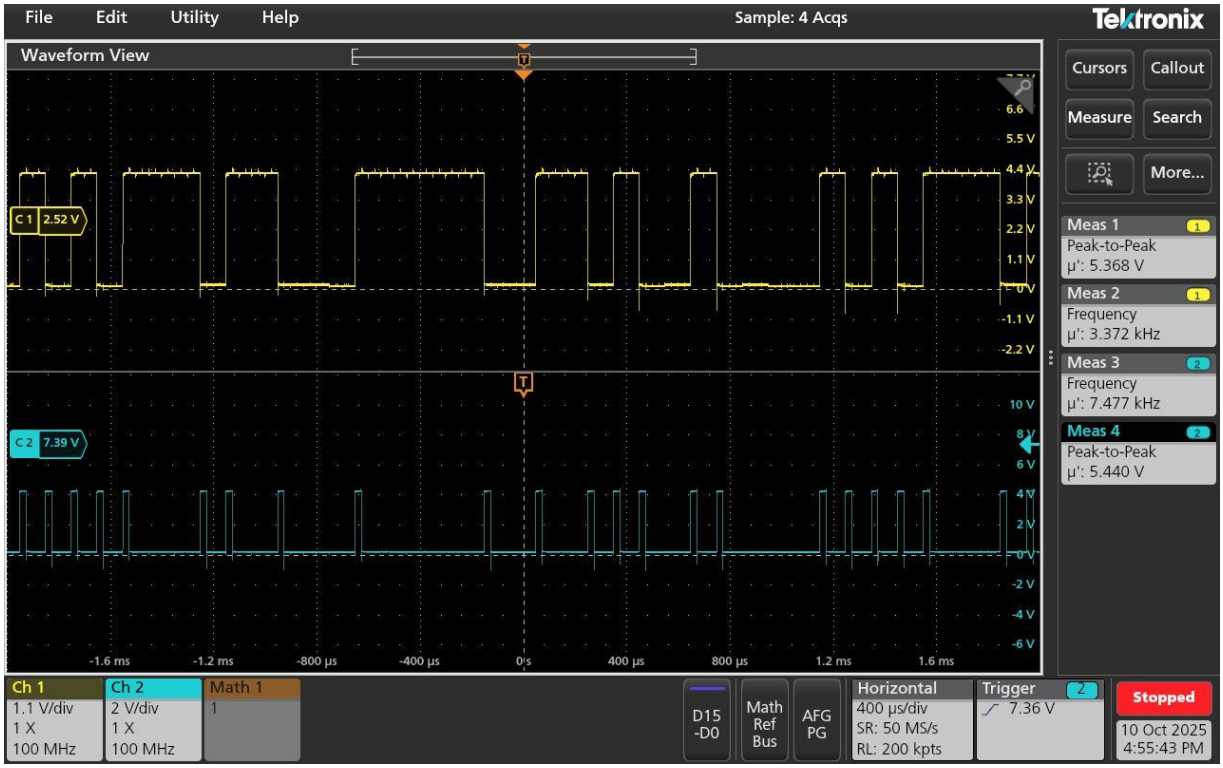


Fig 7: PN Sequence Data generated along with the edge detections after using edge detector circuit

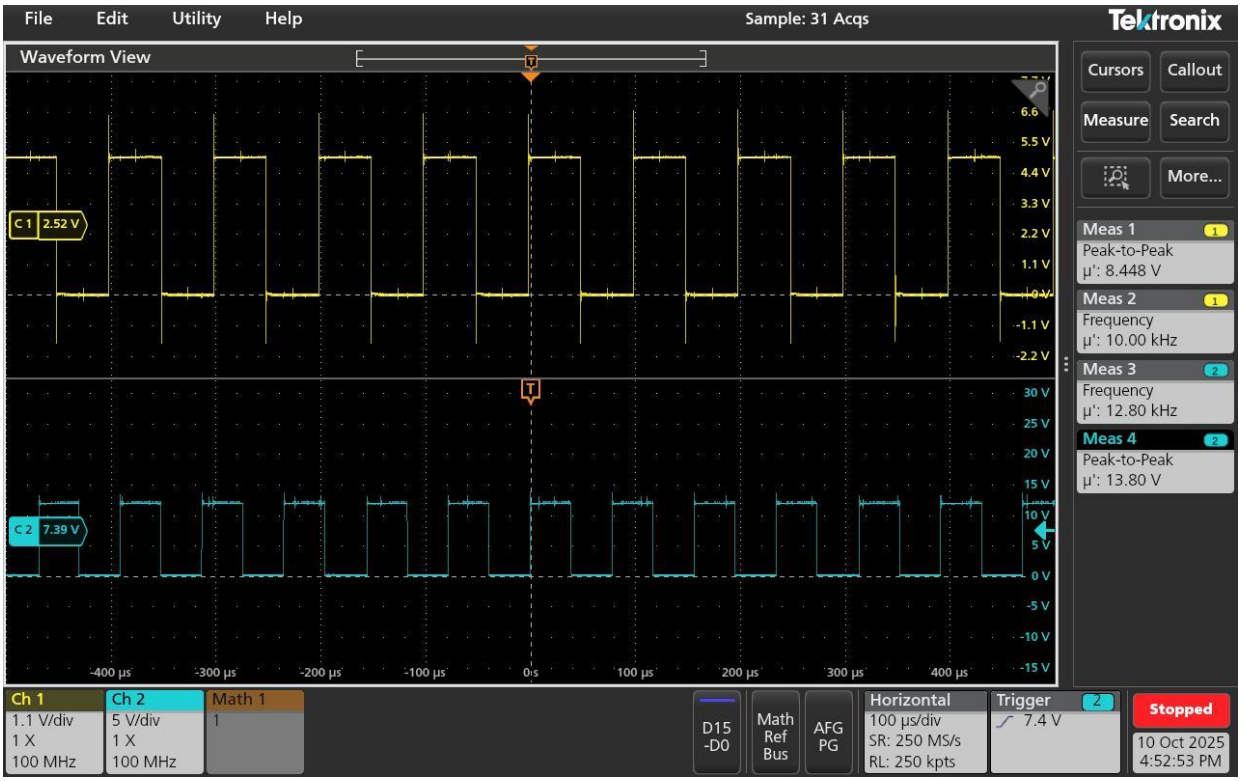


Fig 8: Input Clock data and the clock data recovered after using the circuit

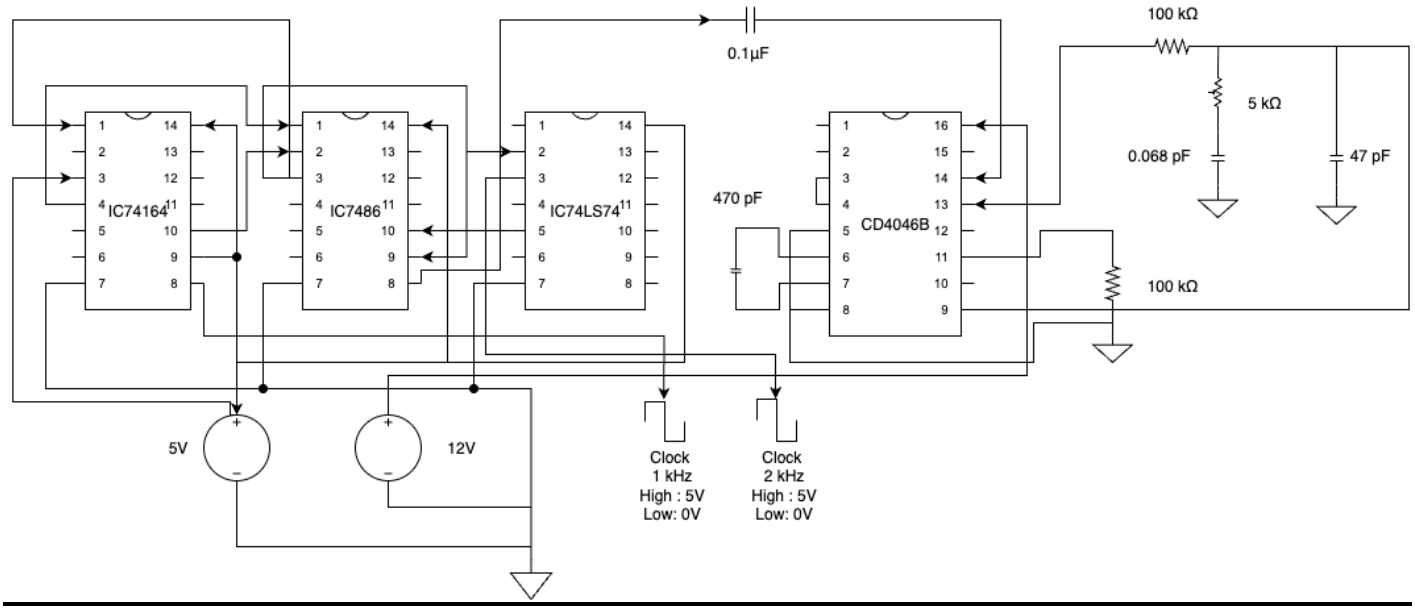


Fig 9: Circuit Diagram showing the ICs along with the components of the loop filter done in laboratory

DISCUSSIONS:

1. The designed circuit has successfully recovered the clock signal from the transmitted PN sequence using the CD4046PLL.
2. The edge detector circuit converted data transitions into short pulses, enabling the PLL to lock onto the correct phase of the data stream.
3. The loop filter played a crucial role in ensuring stability of the feedback loop and controlling jitter.
4. The filter designed corresponds to a second-order lead–lag network, providing one zero and two poles.
5. This configuration offers a good balance between fast lock time and jitter suppression
6. The recovered clock exhibited visible jitter — slight fluctuations in the period and phase of the clock edges when compared with the reference clock.
7. Improvement of the circuit includes the changing of the resistance values (potentiometer hare fails as they were limited to 10kΩ). So multiple resistor and capacitor values can be used to get the result as close as possible to 10kHz (See Group number 13 results).
8. To calculate the $\frac{V_o}{V_{in}}$ value in the loop filter, we need to calculate the overall impedance seen and then calculate the ratio. Following steps do just that:

C_b is grounded with pin 9 connected.

$$Z_{C_b} = \frac{1}{sC_b}$$

R_a and C_a is connected in series with each other and they are connected to ground.

$$Z_{R_aC_a} = R_a + \frac{1}{sC_a}$$

Overall impedance of Z_{C_b} and $Z_{R_aC_a}$:

$$Z_p = \frac{Z_{C_b} Z_{R_aC_a}}{Z_{C_b} + Z_{R_aC_a}}$$

Required $\frac{V_o}{V_{in}}$ ratio:

$$\frac{V_o}{V_{in}} = \frac{Z_p}{R_b + Z_p}$$

Here the system is one zero, two poles (assuming $R_b >> R_a$).

$$\frac{V_o}{V_{in}} = \frac{1 + sR_aC_b}{sR_bC_b(1 + sR_aC_a)}$$

Based on the values of the assumptions, frequencies between 0kHz and 100kHz can be detected.

ACKNOWLEDGEMENTS:

I discussed the findings and the experiment with my lab partner Akash Kumar, my colleagues – Abhijit Mondal, Sikta Mistry, Saptarshi Bhattacharjee. Special thanks to Soumyajit Adak for discussing the loop filter design and how to get the frequency of the clock pulse closer to 10kHz. I am thankful to Prof Amitalok J Budkuley and TA M Jyothi Kiran for looking up the results and suggesting the circuit changes.

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